

Effect of nutrient intake on premenstrual depression.

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We examined the occurrence and coincidence of depressed mood and excessive carbohydrate intake in 19 patients who claimed to suffer from severe premenstrual syndrome and in nine control subjects, all as inpatients, during the early follicular and late luteal phases of their menstrual cycles. Mood was assessed with the Hamilton Depression Scale and an addendum that evaluated fatigue, sociability, appetite, and carbohydrate craving. Calorie and nutrient intakes were measured directly. The subjects with premenstrual syndrome significantly increased calorie intake during the late luteal phase (from 1892 $\pm$ 104 to 2395 $\pm$ 93 kcal, mean $\pm$ SEM); carbohydrate intake increased by 24% from meals and by 43% from snacks. Protein intake failed to change, whereas intake of fat, a fixed constituent of all of the test foods, rose in proportion to calorie intake. The Hamilton Depression Scale and addendum scores rose from 2.0 $\pm$ 0.5 to 21.2 $\pm$ 0.8 (Hamilton Scale) and from 0.5 $\pm$ 0.5 to 10.2 $\pm$ 0.6 (addendum) among subjects with premenstrual syndrome during the luteal phase but failed to change among the controls (2.1 $\pm$ 0.8 to 2.4 $\pm$ 0.8, and 0.4 $\pm$ 0.3 to 0.6 $\pm$ 0.3). Consumption of a carbohydrate-rich, protein-poor evening test meal during the late luteal phase of the menstrual cycle improved depression, tension, anger, confusion, sadness, fatigue, alertness, and calmness scores (p less than 0.01) among patients with premenstrual syndrome. No effect of the meal was observed during the follicular phase or among the control subjects during either phase. Because synthesis of brain serotonin, which is known to be involved in mood and appetite, increases after carbohydrate intake, premenstrual syndrome subjects may overconsume carbohydrates in an attempt to improve their dysphoric mood state.